Policy-Guided Real2Sim2Real for Contact-Rich Robotic Manipulation

Author list and affiliations to be confirmed

Abstract: We ask whether a deployed policy's behavior can calibrate a reconstructed, interactive simulator that then improves the same policy. On plug insertion with a frozen $\pi_{0.5}$ checkpoint, the policy's first action exposes visual mismatch that appearance hides, and a bounded residual trained in the reconstructed simulator raises insertion from 7/20 to 13/20 in simulation and from 6/20 to 11/20 on the real robot ($p = 0.20$). The uncalibrated-simulator control needed to attribute this gain to calibration has not been run.

Keywords: Real-to-sim-to-real, simulator calibration, residual policy learning

1 Problem and claim

Plug insertion fails on millimetre errors in contact geometry, alignment and appearance that a convincing digital twin can still get wrong. Simulation evaluation [1], digital-twin learning [2] and agentic reconstruction [3] make twins cheaper, but do not ask which mismatches matter to the policy being trained. Hypothesis: a simulator calibrated to reproduce a deployed policy's behavior is a better environment for improving that policy than one calibrated for appearance alone (Figure 1); we report which links of this chain are measured.



Figure 1: Policy-guided Real2Sim2Real for plug insertion. Geometry and appearance come from the image; socket cavities and other task contacts are authored; camera and appearance are tuned against frozen-policy responses. The dashed validation gate is a proposed stopping rule, not an achieved convergence result.

2 Method

Interactive reconstruction. A SimFoundry-derived pipeline [4] turns one RGB image into a settled scene (metric depth, meshes, FoundationPose poses, convex colliders); a harness gates each stage and rejects runs with missing or misregistered objects or more than 3 cm of settling. Images cannot supply task contacts, so the reconstructed strip keeps its body and

texture but receives authored socket cavities. Quantities are thus image-derived (geometry, pose), authored (contacts) or policy-tuned (camera, appearance).

Policy-guided calibration. For a frozen policy π we seek simulator parameters $\theta^* = \arg\min_\theta D\big(\tau_{\text{real}}(\pi), \tau_{\text{sim}}(\pi;\theta)\big)$, where θ holds camera extrinsics, colour response and surface appearance, and D compares action responses at corresponding states with proprioception, prompt and sampling fixed. We evaluate D as the fraction of home-state samples whose first action chunk descends, and search θ by agent-assisted edits: there is no automatic optimizer, and contact parameters are not searched.

Residual policy. The base $\pi_{0.5}$ checkpoint 14999 stays frozen. An MLP reads its prefix features, frozen DINOv2 features of both cameras and joint state, and outputs a fingertip shift Δ_{xy} clipped to ± 20 mm and a gate g_ϕ; one damped least-squares step adds it to every row of the base chunk, $q_{1:50} \leftarrow q^{\text{base}}_{1:50} + J^\dagger[\Delta_{xy}, 0]$, when $g_\phi > 0$ and the gripper holds the plug. Labels come from a privileged oracle that takes over during insertion in 80 simulated episodes (seeds disjoint from evaluation); the loss is smooth-L_1 on Δ_{xy} plus $0.3\times$ cross-entropy on the gate. This is supervised residual imitation, not residual RL [5], and no object pose reaches the head.

3 Results

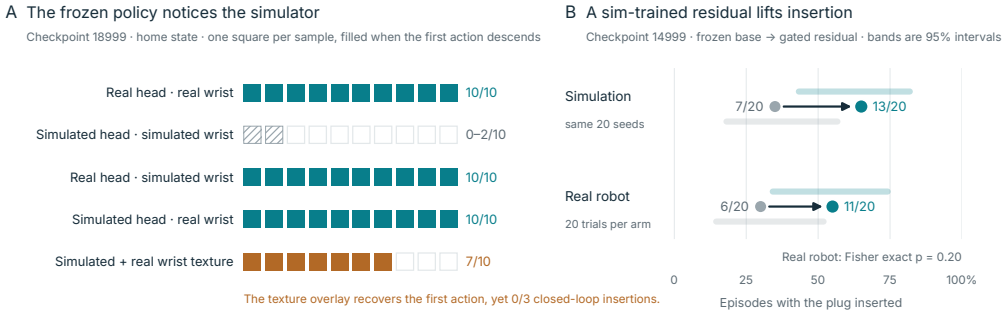


Figure 2: A: Frozen checkpoint 18999 at the home state, $n = 10$ samples per camera condition. B: Frozen checkpoint 14999 with and without the gated residual: simulation on the same 20 seeds, and 20 real-robot trials per arm. Wilson 95% intervals.

The policy detects mismatch that looks minor. Checkpoint 18999 descends in 10/10 home-state samples with real images but 0–2/10 with simulated ones; swapping in either real camera restores 10/10. Real wrist texture gives 7/10 yet 0/3 closed-loop insertions, so matching the first action does not establish closed-loop fidelity.

Contact mechanisms behave as intended. A three-prong probe seats to 15.9 mm and holds for 2 s, and is rejected at 90°/180° rotation and a 4 mm offset (fixture tests; the task uses a two-prong charger).

The residual improves insertion. On the same 20 simulation seeds, insertion rises from 7/20 (frozen base) to 13/20 (gated residual); on the real robot, from 6/20 to 11/20 (Fisher exact $p = 0.20$). The direction agrees across sim and hardware; 20 trials per arm cannot fix the effect size.

What is supported, and what is not. Supported: the frozen policy probes visual mismatch; authored contacts make a reconstructed asset usable for insertion; a residual trained only in simulation transfers with a positive trend. Not yet measured: a discrepancy-reduction curve for an automatic search; a residual trained in an uncalibrated simulator, the control that would attribute the real gain to calibration; held-out real states; and identification of contact parameters such as friction and compliance. Next: base vs. residual from an uncalibrated vs. a policy-calibrated simulator, on the real robot from fixed starts.

References

[1] X. Li, K. Hsu, J. Gu, K. Pertsch, O. Mees, et al. Evaluating real-world robot manipulation policies in simulation. In Conference on Robot Learning, 2024. URL https://simpler-env.github.io/.

[2] M. Torne, A. Simeonov, Z. Li, A. Chan, T. Chen, A. Gupta, and P. Agrawal. Reconciling reality through simulation: A real-to-sim-to-real approach for robust manipulation. In Robotics: Science and Systems, 2024. URL https://arxiv.org/abs/2403.03949.

[3] R. Luo, Y. Liu, D. Cao, M. Tancik, S. Buch, C. Stearns, and W. Xian. SceneRig: An agentic system for simulation-ready 3d scene reconstruction from single images. arXiv preprint arXiv:2610.28585, 2026. URL https://arxiv.org/abs/2610.28585.

[4] NVIDIA. SimFoundry: Modular and automated scene generation for policy learning and evaluation. Software repository, 2026. URL https://github.com/NVlabs/SimFoundry.

[5] K. Kim, N. Saito, H. Kim, K. Ikeuchi, J. Choo, and Y. Matsushita. Object-centric residual RL for zero-shot sim-to-real VLA enhancement. arXiv preprint arXiv:2606.18953, 2026. URL https://arxiv.org/abs/2606.18953.